

Image



Image



Title

The Internet of Cars: A Business Case

Page-text

Intelligent Transportation Systems World Congress 2011

Image



Image



Image



Other-text

Andreas Mai
Director Automotive North America
Internet Business Solutions Group

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Image



Other-text

18 October 2011

Image



Image



Transportation: A Global Transformational Opportunity?

Diagram

Obj-text

Accidents

Obj-text

- 7.9 million accidents

Obj-text

- 7.4 million injured

Obj-text

- 1.3 million fatalities

Obj-text

- \$0.6 trillion cost

Image



Obj-text

Congestion

Obj-text

- 90 billion hours delay

Obj-text

- 230 million metric tons of carbon-equivalent

Obj-text

- \$1 trillion waste (2% of GDP)

Obj-text

Future of Mobility?

Image



Obj-text

Auto Business Model

Obj-text

- Oversized and underutilized vehicles

Obj-text

- 75% of automotive value add post sale

Obj-text

- 8%-15% gross margins

Obj-text

Road/ Traffic System

Obj-text

- "Free" public service

Obj-text

- No/ low return on assets

Obj-text

- Underfunded

Obj-text

- Deficient

Other-text

Source: IRF World Road Statistics 2008, World Bank Global Road Safety Facility, Cisco IBSG Automotive 2011

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U.S. Roads and Highways Are Congested, Deficient, Lack Desirable Safety Features

Diagram



Obj-text
25%

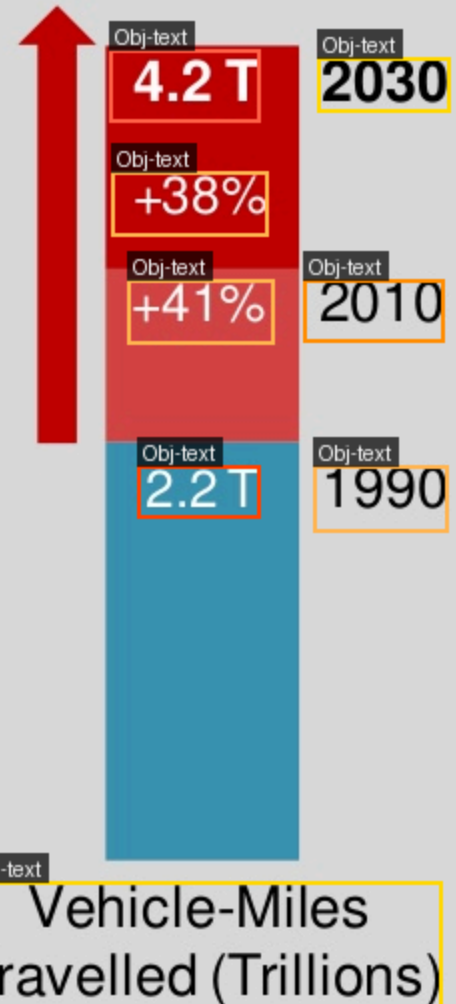
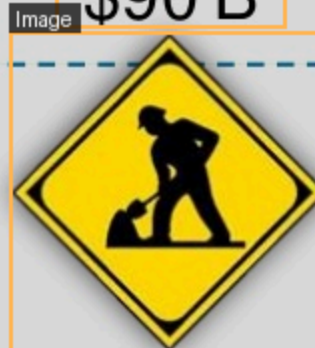


Obj-text
\$270 B
Needed



Obj-text
Cost
+ 55%

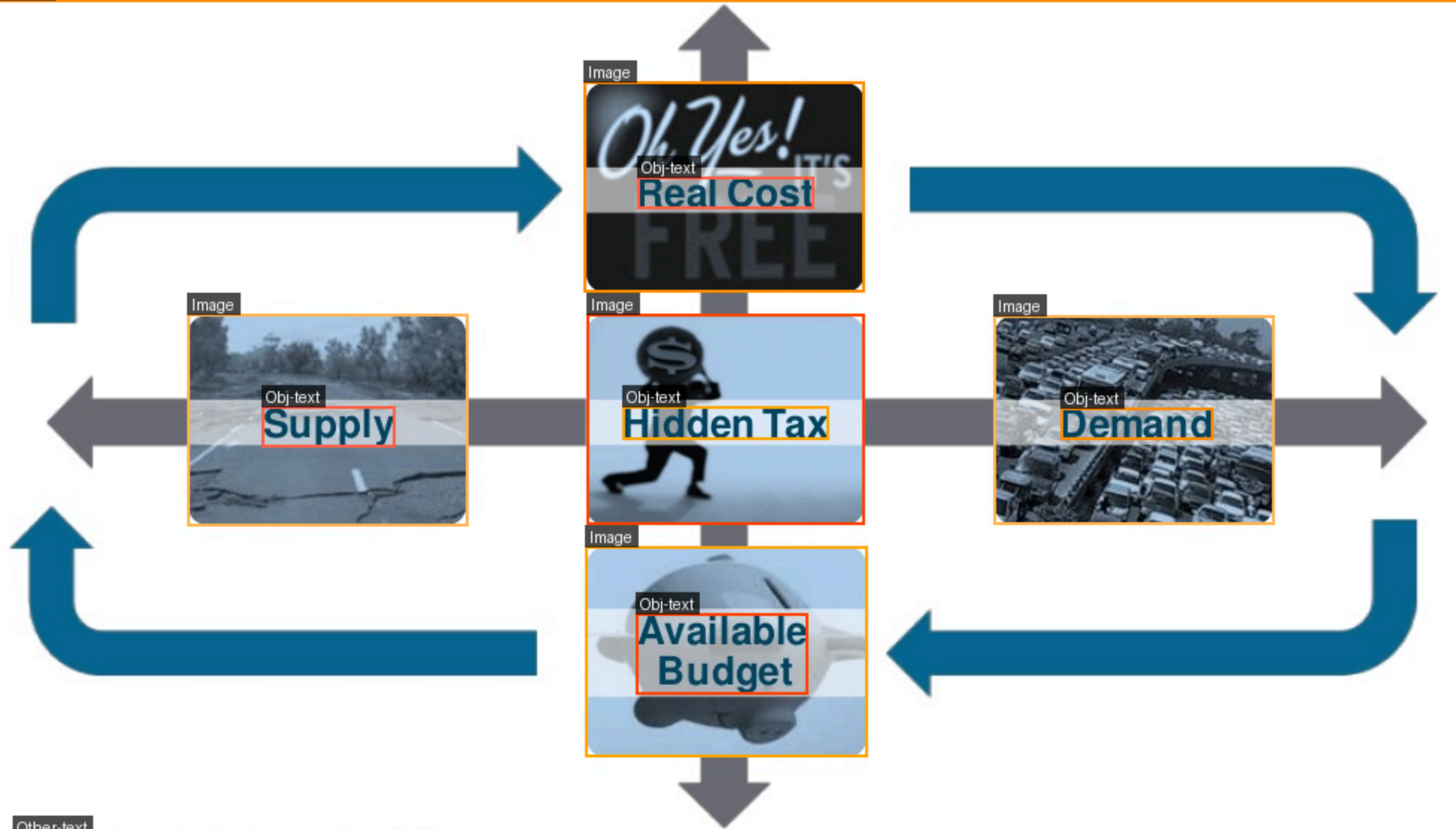
Obj-text
Budget:
\$90 B



Other-text
Source: America's Top Five Transportation Headaches, AASHTO, Victoria Transportation Policy Institute, 2009

Vicious Circle: Pigouvian Tax for Personal Mobility

Diagram



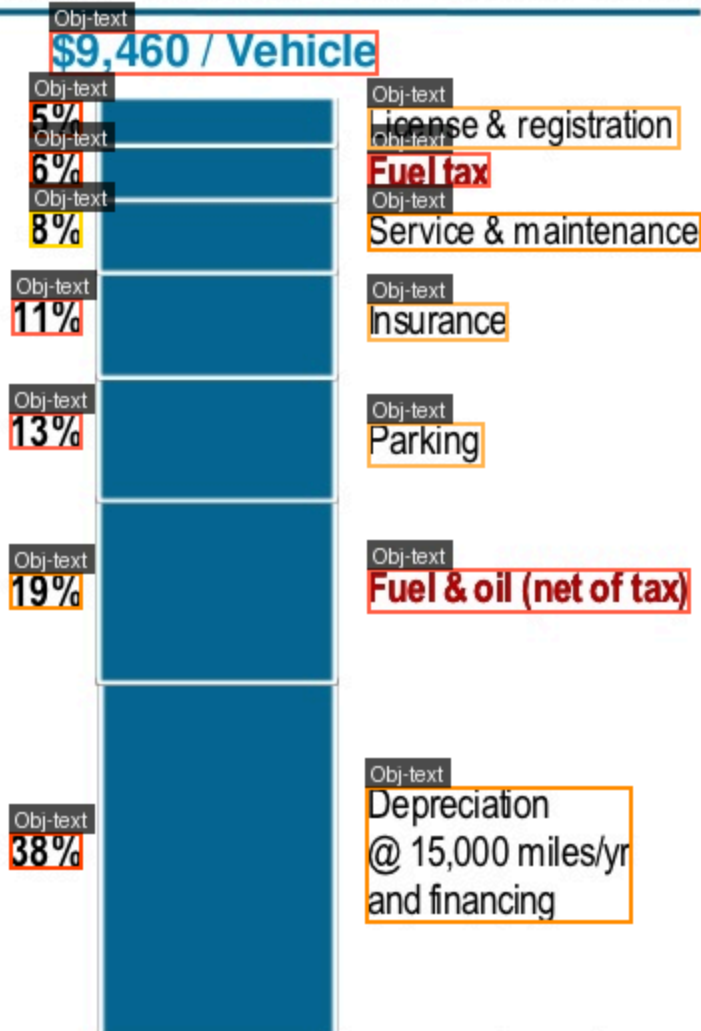
Source: Cisco IBSG Automotive, 2011

“Real” Cost of Owned Vehicles in U.S. Is \$3 Trillion Per Year—50% Is Hidden Tax

Diagram

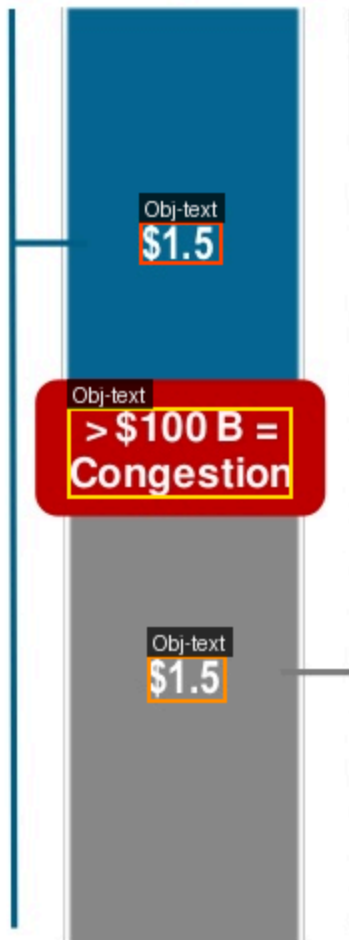
Caption

“Visible” Annual U.S. Vehicle Cost



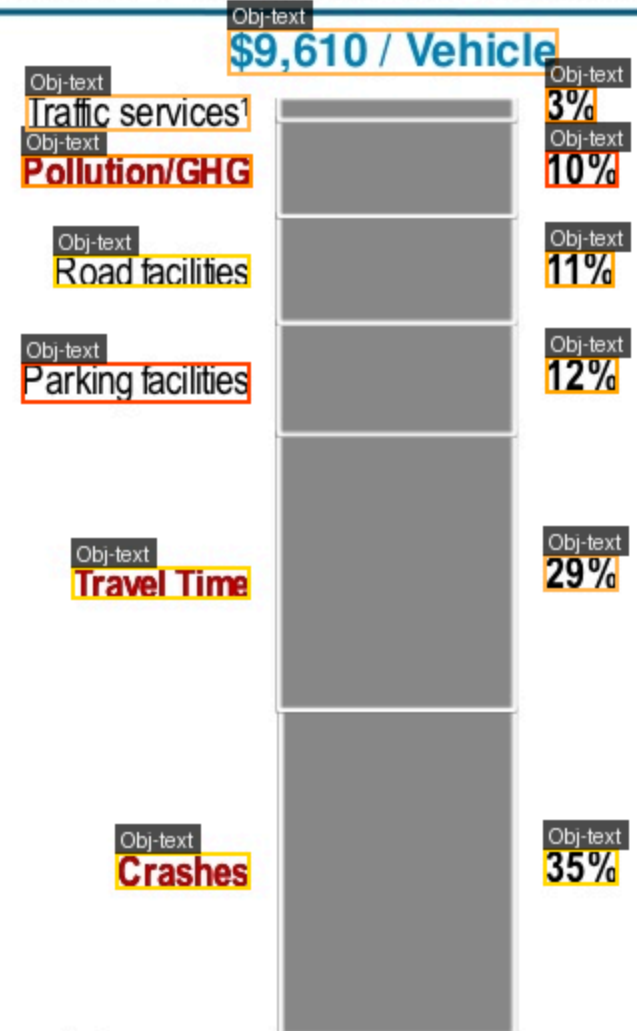
Obj-text

\$3 Trillion / Year



Caption

“Hidden”/ Societal U.S. Traffic Tax



Other-text

Source: Victoria Transport Policy Institute, AAA, DOT, Cisco IBSG Automotive 2011

Other-text

1. Includes toll, law enforcement

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Title

In Urban Traffic, Even the Most Fuel Efficient Car Wastes Energy and Time

Caption

Urban Congestion:

Obj-text

11%-13% Time Wasted

Image

Obj-text

1 Liter / 100 km Car



Image



Other-text

Source: VDA Annual Report, TTI, VTPI, Cisco IBSG 2011

Caption

Parking Search:

Obj-text

7%-21% of Urban Traffic

Image



Image



Caption

Traffic Light Stops:

Obj-text

10%-17% of Urban Fuel

Image



Image



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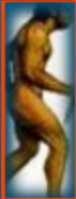
Evolution of Ground Transportation?

Diagram

Caption

Personal Transportation

Image



Image



Image



Image



Image



Image



Image



Obj-text

...But What is the Next Big Thing?

Caption

Mass Transportation

Image



Image



Image



Image



Image



Other-text

Source: christian-foerg.de ,Cisco IBSG Automotive 2011

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Reinventing Ground Transportation

Diagram

Image



Obj-text

Smart

Image



Obj-text

Intelligent Transportation

Obj-text

Networked

Obj-text

Pervasive Computing

Obj-text

Electric

Obj-text

Smart Grid

Other-text
Source: Cisco IBSG Automotive 2011

Other-text

www.cisco.com/ibsg/automotive/2011/01/11/ibsg_automotive_2011

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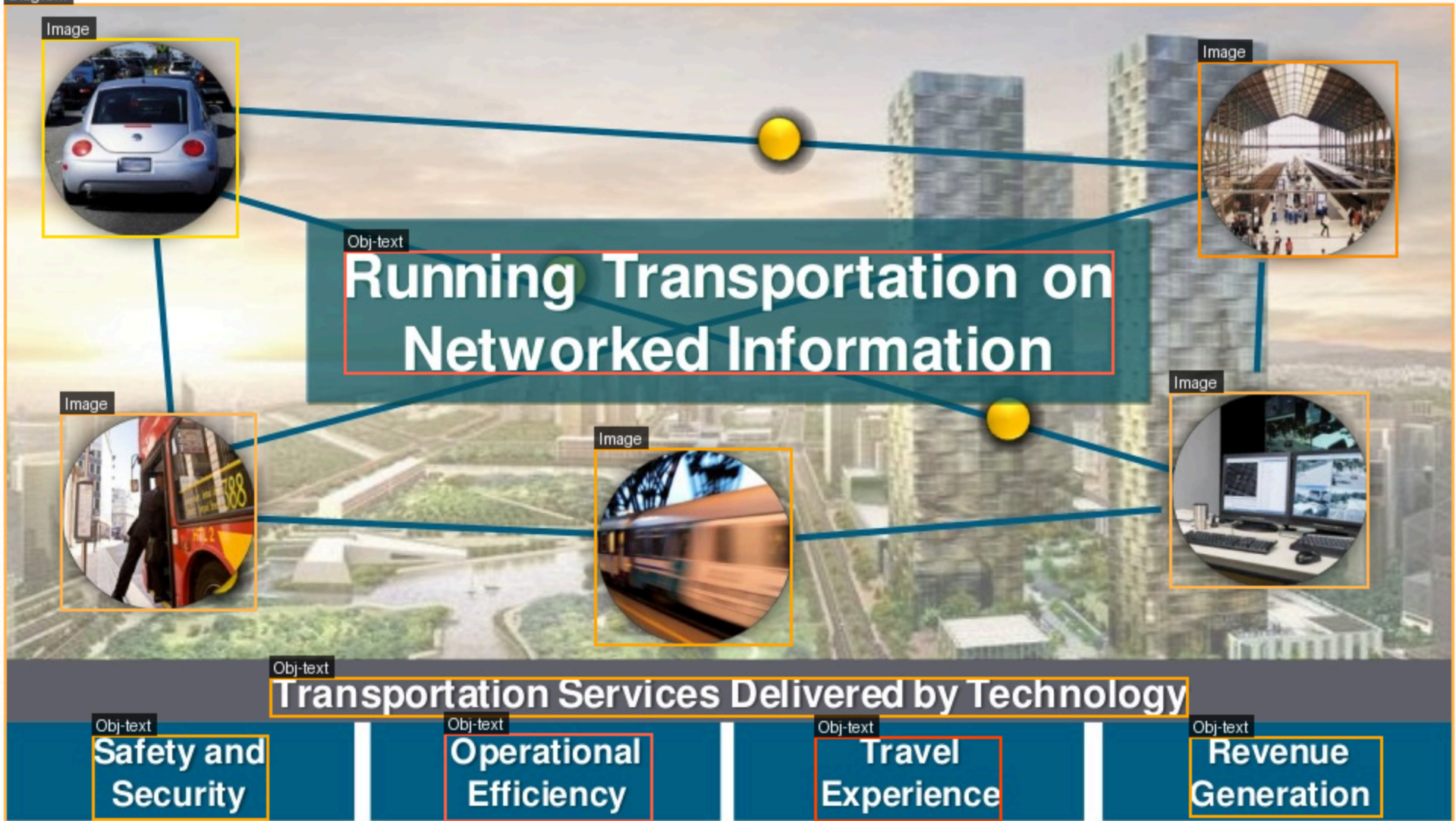
Other-text

8

Title

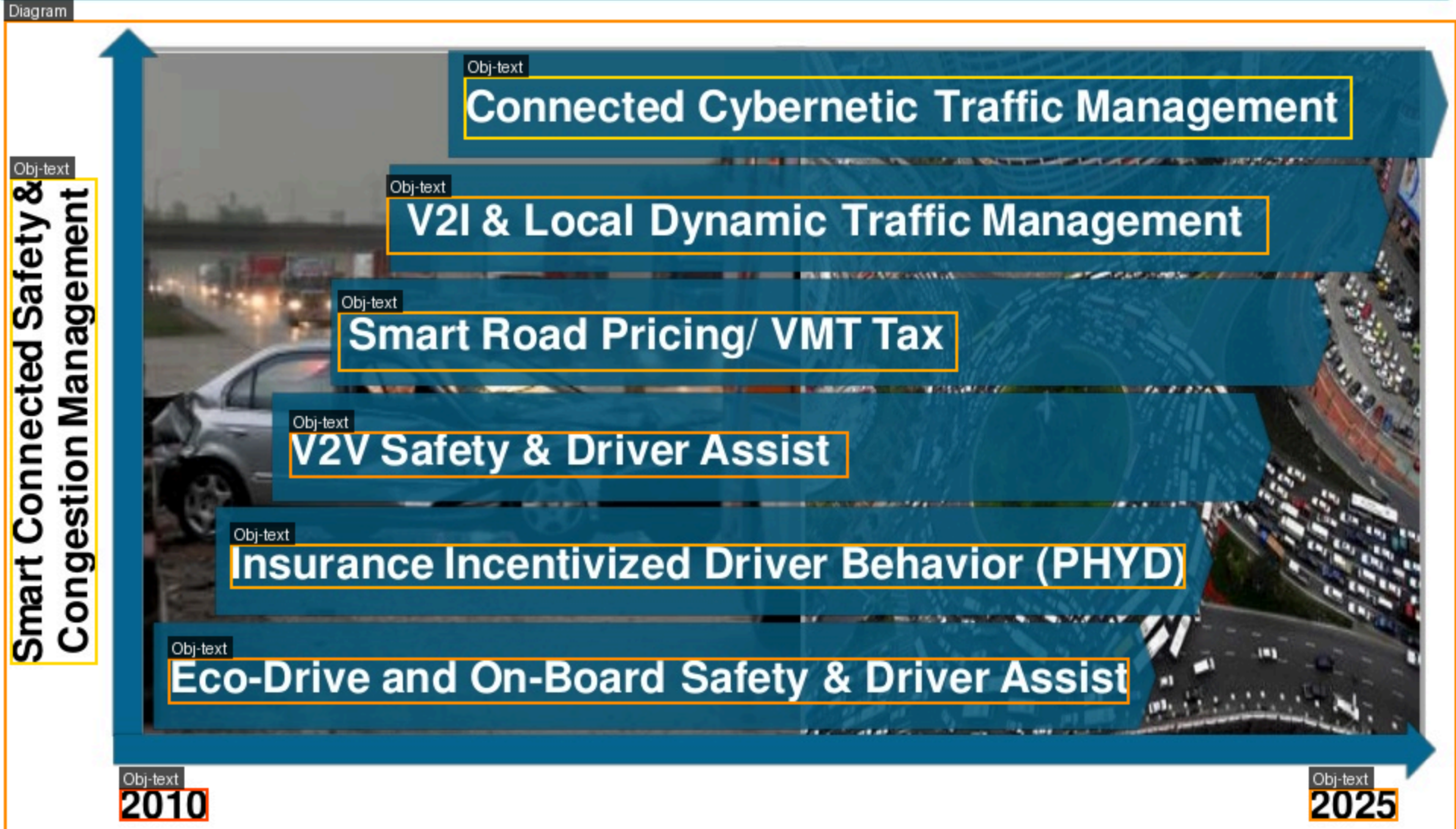
The Internet Takes to the Road...

Diagram



Other-text
Source: Cisco IBSG Automotive, 2011

Road to More Safety & Less Congestion Is Mainly Driven by Government Policy



Source: Cisco IBSG Automotive 2011

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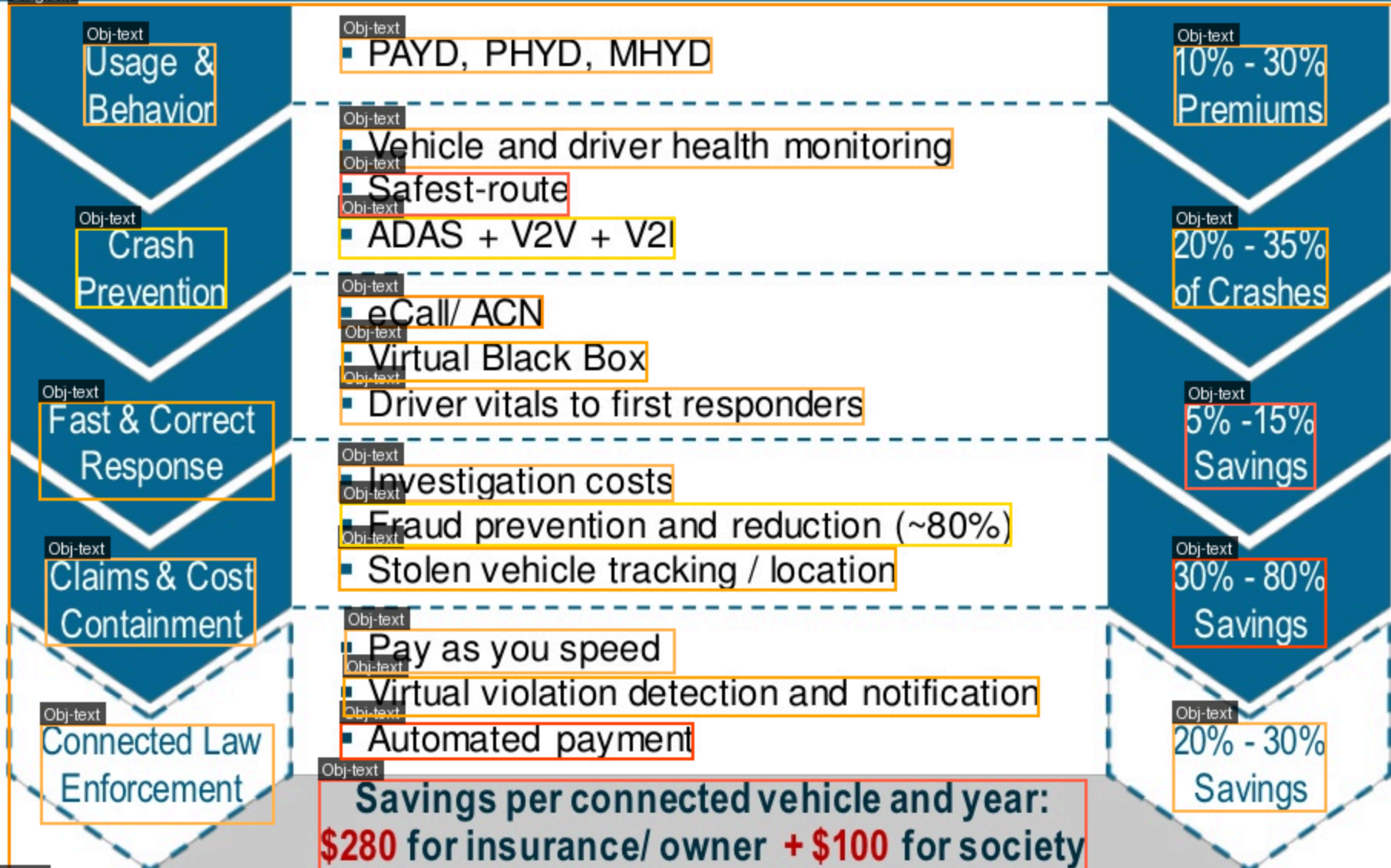


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THE “Savior-App”: Strategic Asset for the Entire Car Insurance Value Chain

Diagram

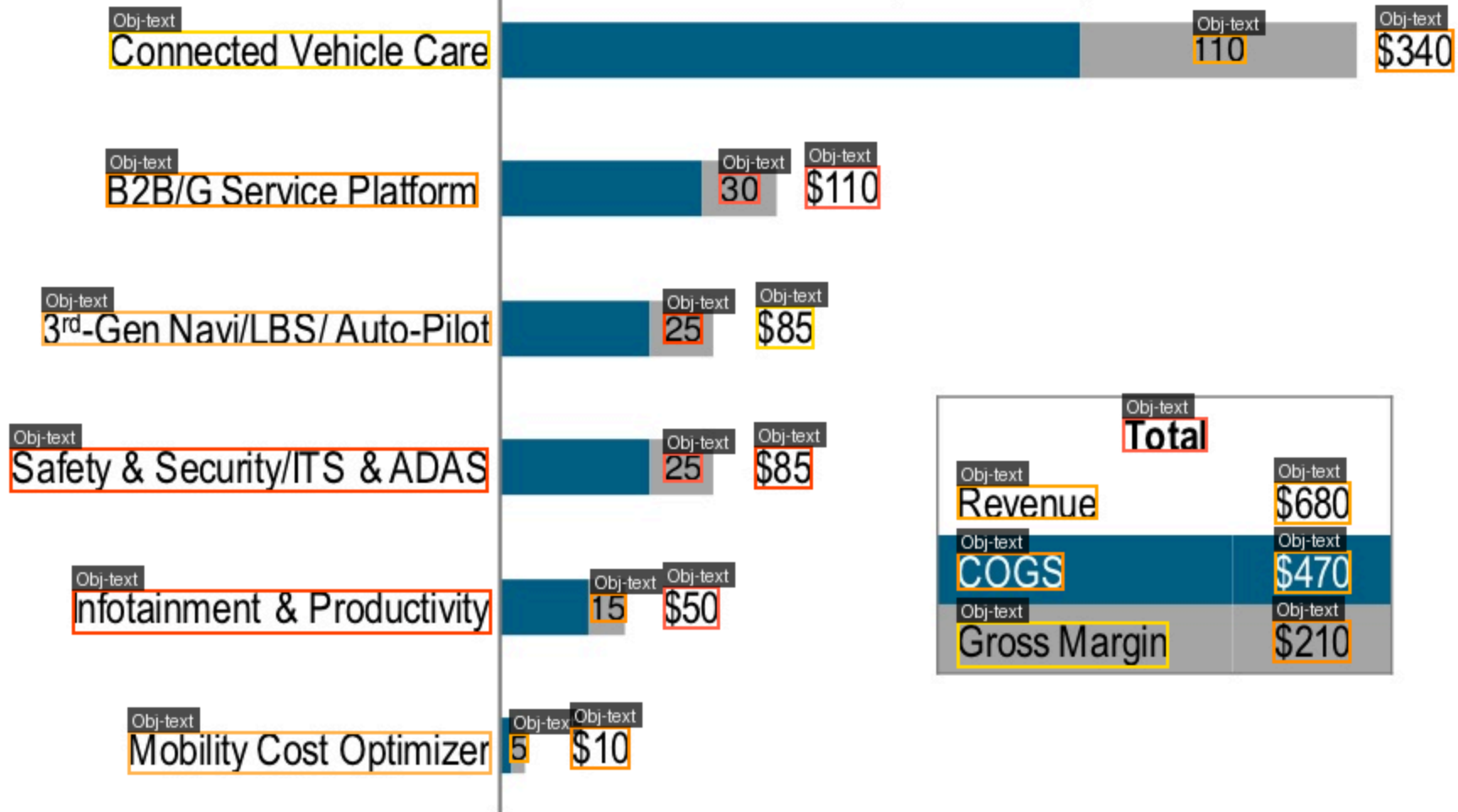


Source: OctoTelematics, Cisco IBSG Automotive 2010

Top Auto Profit Pools Are Connected Vehicle Care and B2B/G Services

Figure

Dollars per Vehicle per Year¹



Includes one-time effects spread over assumed 5-year first ownership cycle
Source: Cisco IBSG Automotive, 2011

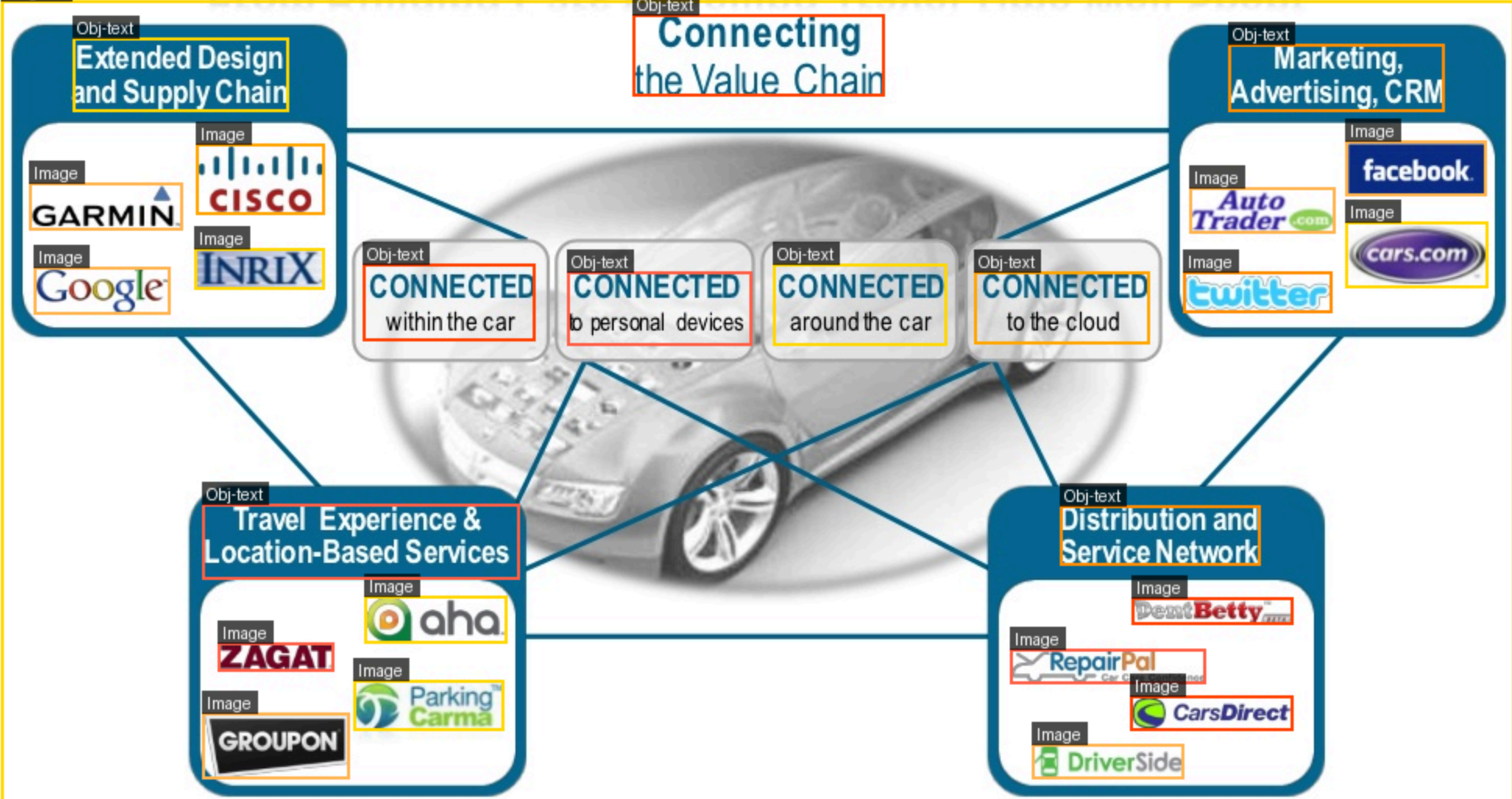
Title

Advent of New Business Models?

Caption

From Building Cars to Selling Travel Time Well Spent

Diagram



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Source: Cisco IBSG Automotive, 2011

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Connecting the Mobility Value Chain: New Business Model Opportunities

Diagram



Source: Cisco IBSG Automotive 2011

Title

Internet of Cars = Better Experience = Key Growth Market for Service Providers

Diagram

Obj-text

Pervasive Computing

Obj-text

Users

Obj-text

Grow the number of mobile revenue-generating subscriptions

Image



Image



Image



Obj-text

Ubiquitous Connectivity

Obj-text

Artificial Intelligence

Obj-text

Uses

Obj-text

Find more things users can do with their mobile devices

Obj-text

M2M Network

Obj-text

Usage

Obj-text

Drive the frequency with which people use mobile data services

Other-text

Source: Cisco IBSG 2010

Other-text

www.cisco.com/c/en/us/solutions/industry/transportation/connected-vehicles/index.html

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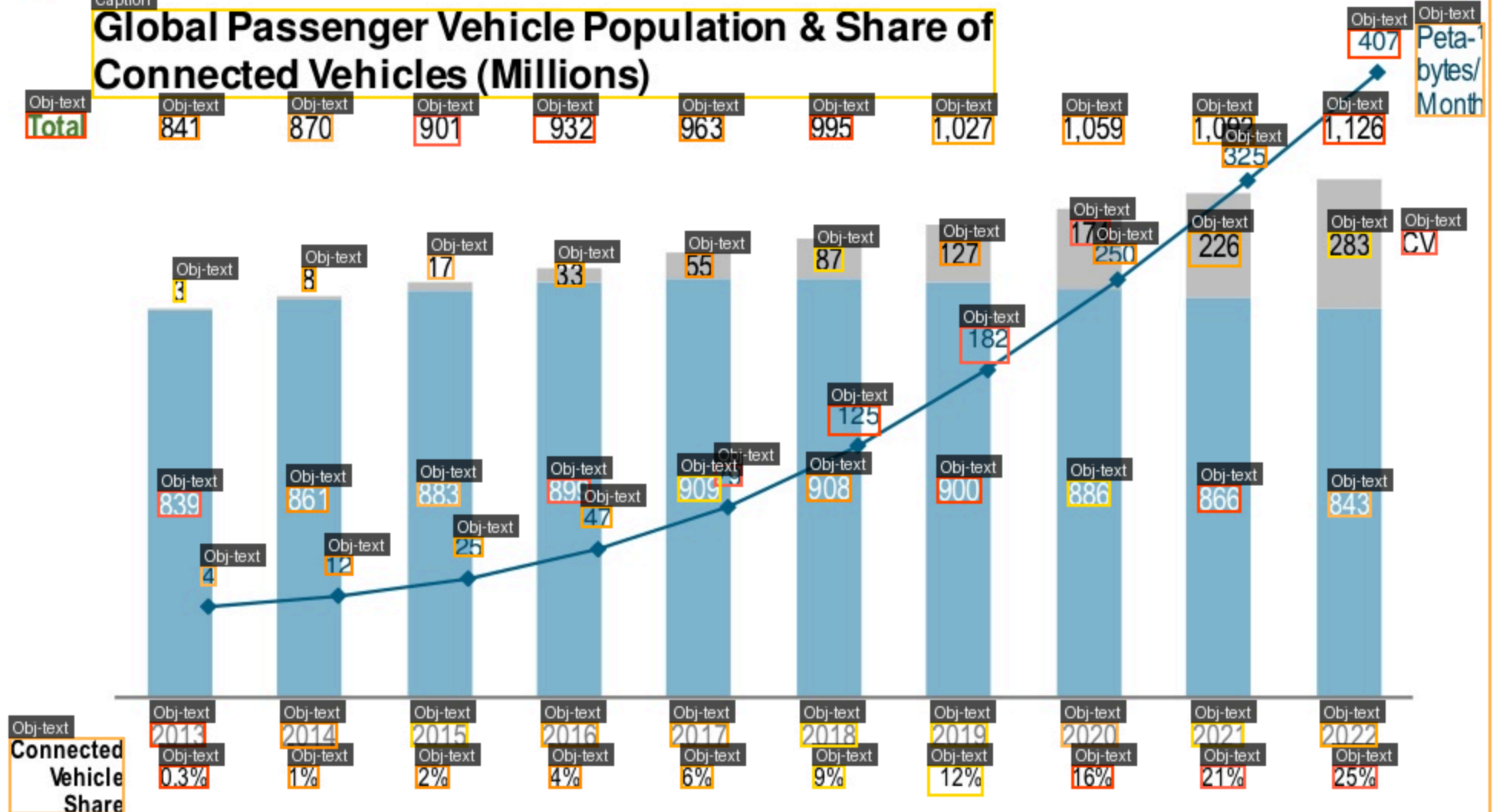
18

Nearly 300 Million Connected Vehicles = More Than 400 Million Gigabytes / Month

Figure

Caption

Global Passenger Vehicle Population & Share of Connected Vehicles (Millions)

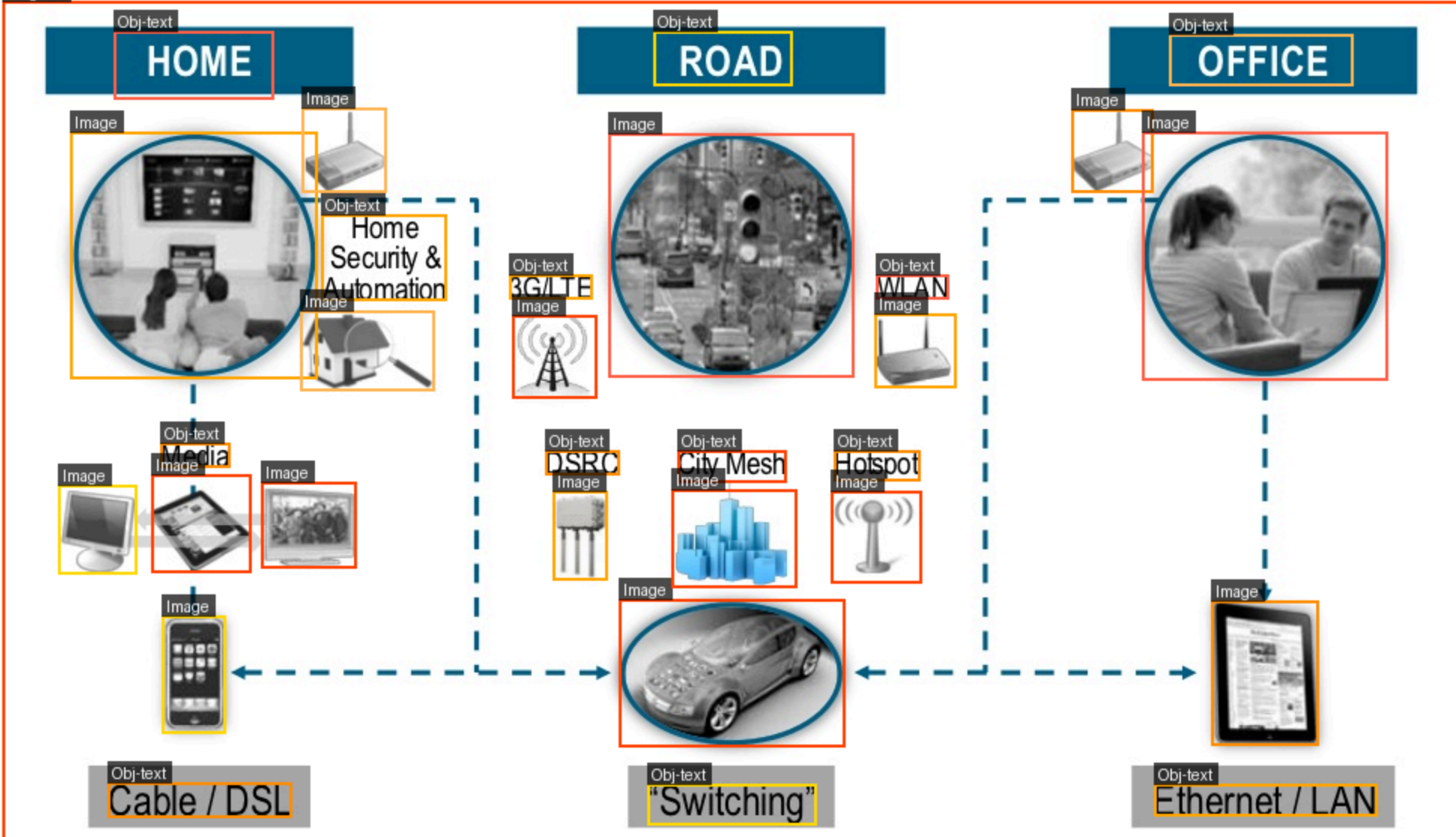


Average of 1.5 GB/month/vehicle, 1 Petabyte = 1,048,576 GB

Source: Cisco IBSG estimate 2011, based on data from US-DOT, iSupply, McKinsey

Off-Loading Data and Switching Is Key to Improve Experience and Contain Cost

Diagram



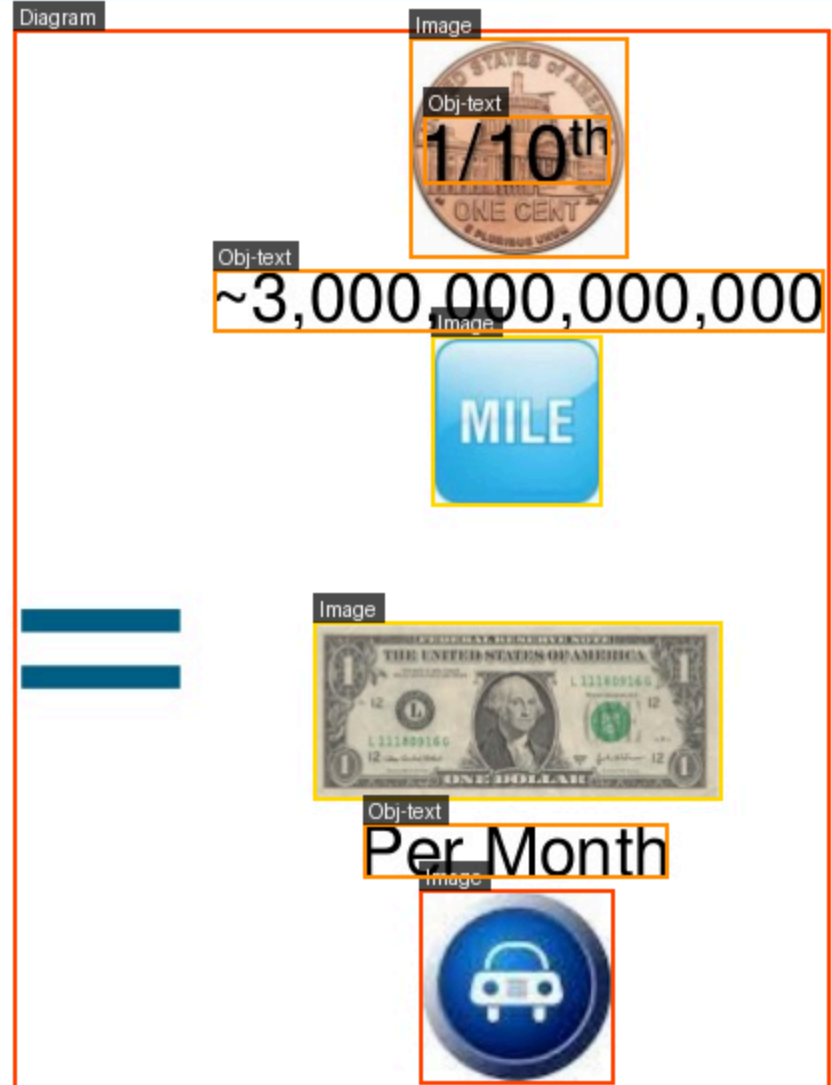
Source: Cisco IBSG Automotive 2011

The Connectivity Infrastructure Could be a Profitable Business at 1/10th ¢ / VMT

U.S. Roadside Equipment (RSE) PPP Revenue Model [US\$ Million]

	CapEx	P&L
Initial Total 5 Year RSE Investment (10 Depr.)	6,261	626
Repair/ Replacement	616	62
Operating Expenses		440
Subtotal RSE Cost		1,128
Hypothetical Margin	100%	1,128
Revenue Required to Fund "RSE Business"		2,256

Source: Vehicle-Infrastructure Integration (VII) Initiative, Benefit-Cost Analysis, Version 2.3, US-DOT, 2008
Cisco IBSG Automotive, 2011



THE Future “Finance-Application”: Connecting for Smart Road Pricing

Caption

Smart Road Pricing / Vehicle-Miles-Traveled Tax

Image



Caption

Time of Day

Image



Caption

Occupancy

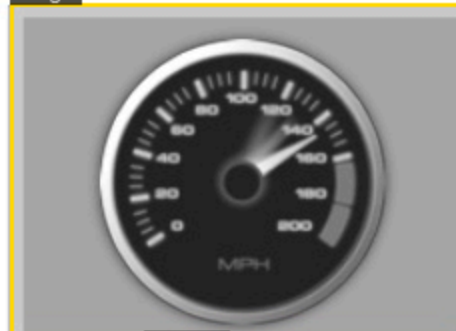
Image



Caption

Congestion

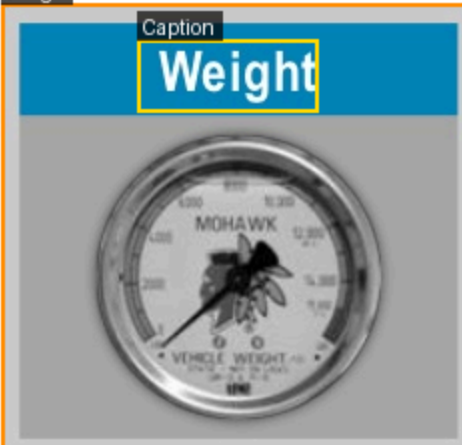
Image



Caption

Speed

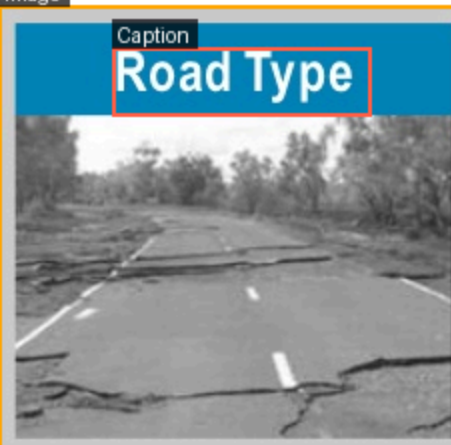
Image



Caption

Weight

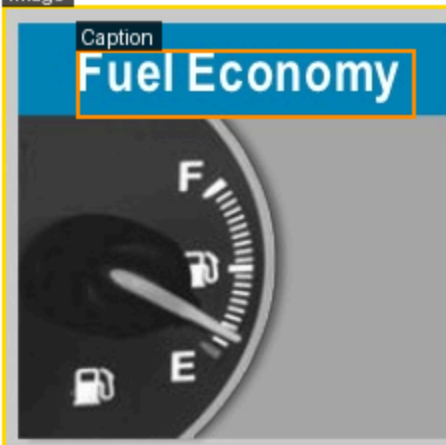
Image



Caption

Road Type

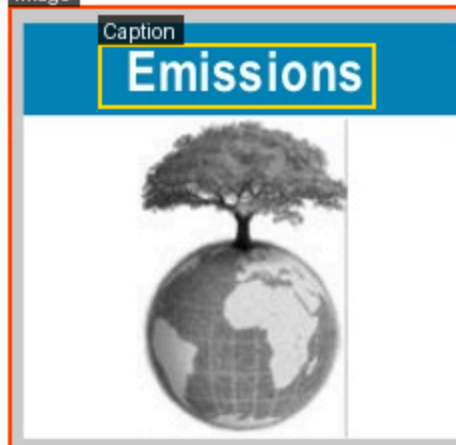
Image



Caption

Fuel Economy

Image



Caption

Emissions

